Measurement-Efficient Port-Reading Policies for Fluid Antenna Systems Under Spatially Correlated $\kappa-\mu$ Shadowed Fading

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Presentation Outline

- Fundamentals: FAS, Shadowed Fading, & the Port-Reading Problem
- ML-Based Channel Prediction from Sparse Measurements
- Synergy: How Optimized Reading Patterns Enhance ML Predictor Accuracy and Efficiency
- Core Contribution & Evaluation

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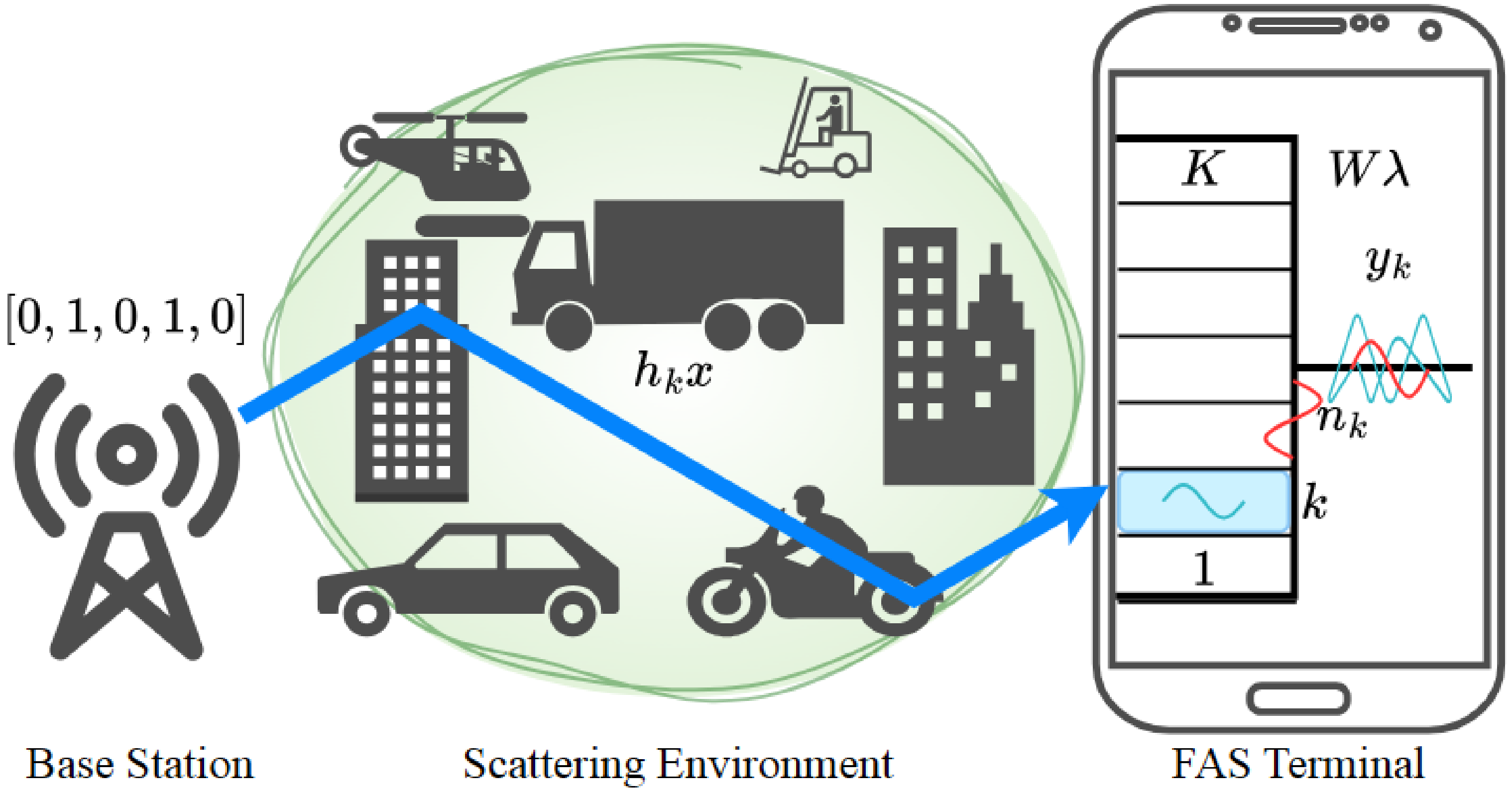
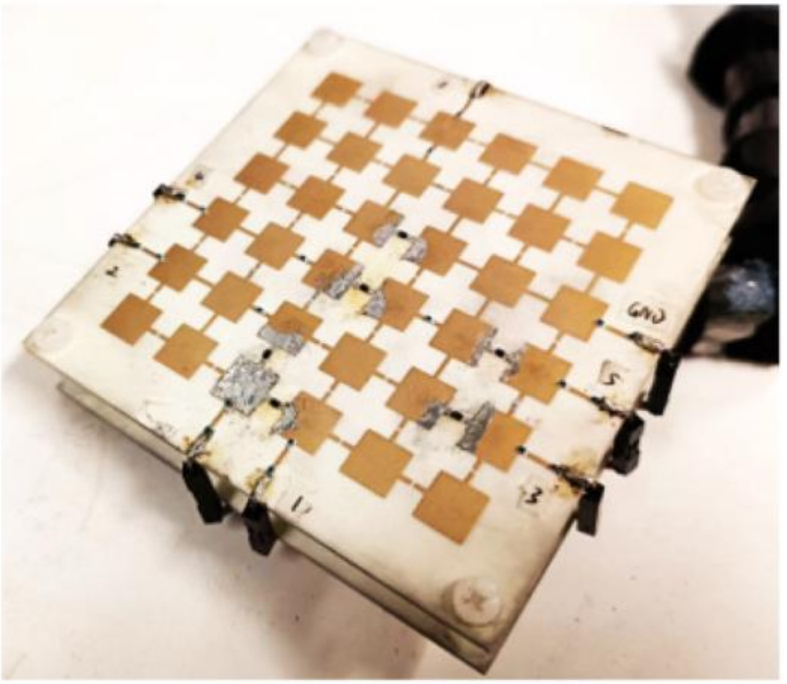
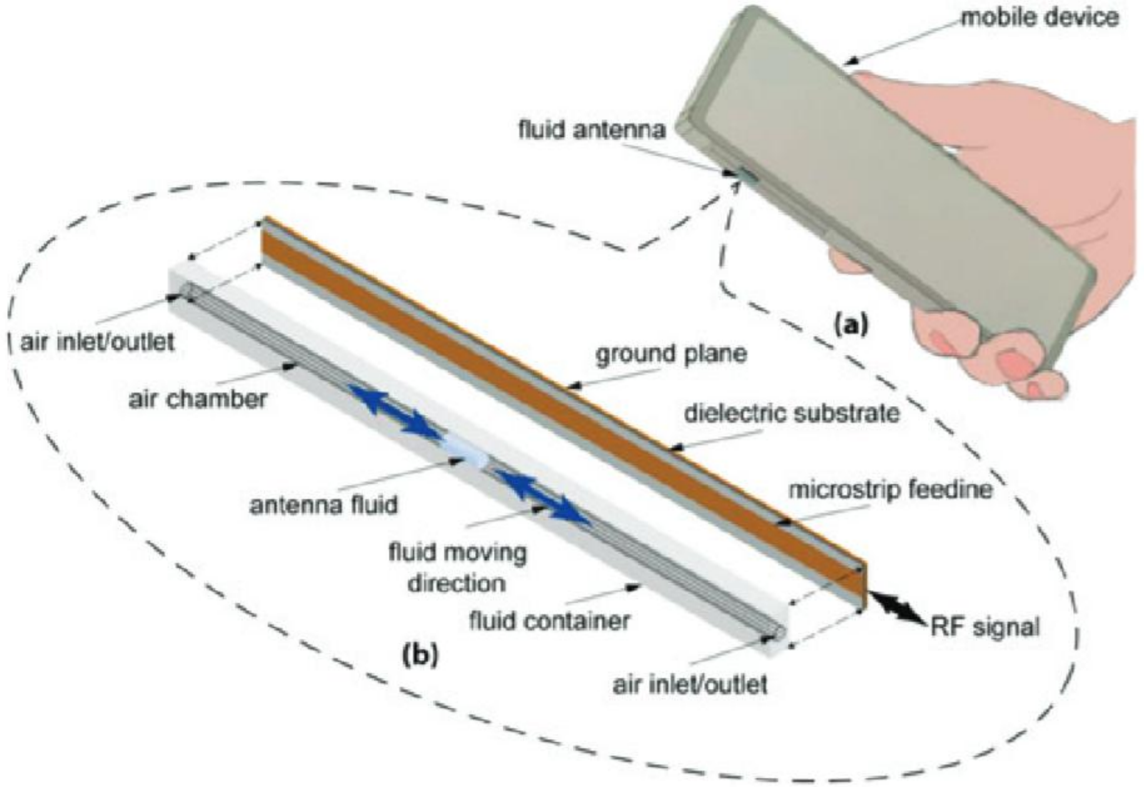
Fluid Antenna System

- The concept of the **Fluid Antenna System (FAS)** emerged to improve performance.
- (2020) Kai Kit Wong





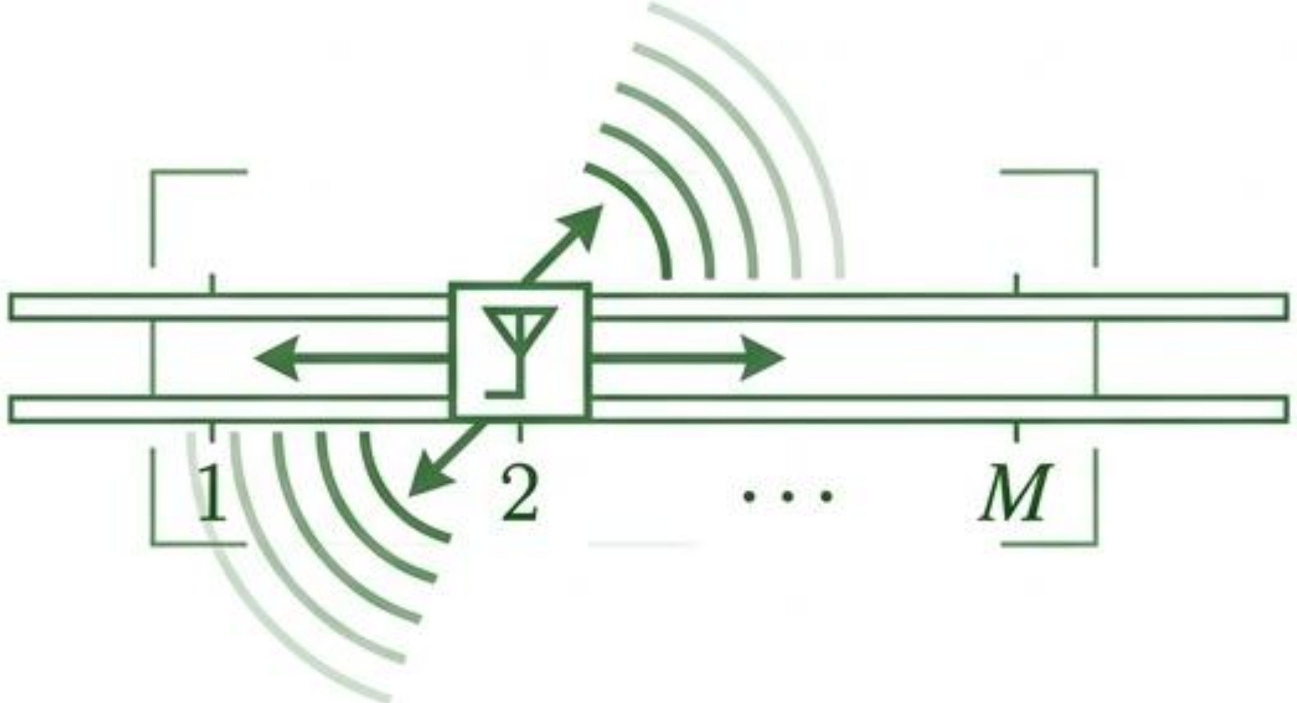
Antenna relocation can be achieved by mechanically moving liquid conductors through a microfluidic system, or by electronically switching radiating pixels on and off.



Low-Power AI-Based Port Selection for Fluid Antennas in Spatially Correlated Channels, WCNPS 2025.
Performance limits of fluid antenna systems, *IEEE Communications Letters*, Nov. 2020.
Fluid antenna system for 6G: When Bruce Lee inspires wireless communications, *Electronics Letters*, Nov. 2020.

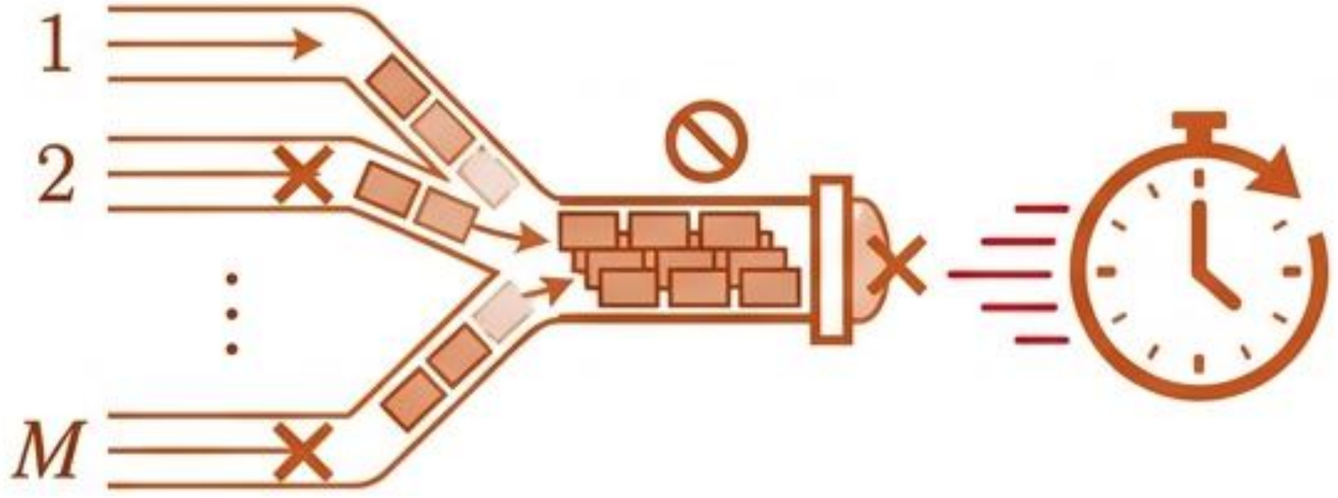
Fluid Antenna Systems (FAS): Massive Diversity vs. Massive Overhead

The Promise



A single RF chain electronically moves among M candidate positions within a small space. Approaches or exceeds classical multi-antenna combining gains without the massive hardware footprint.

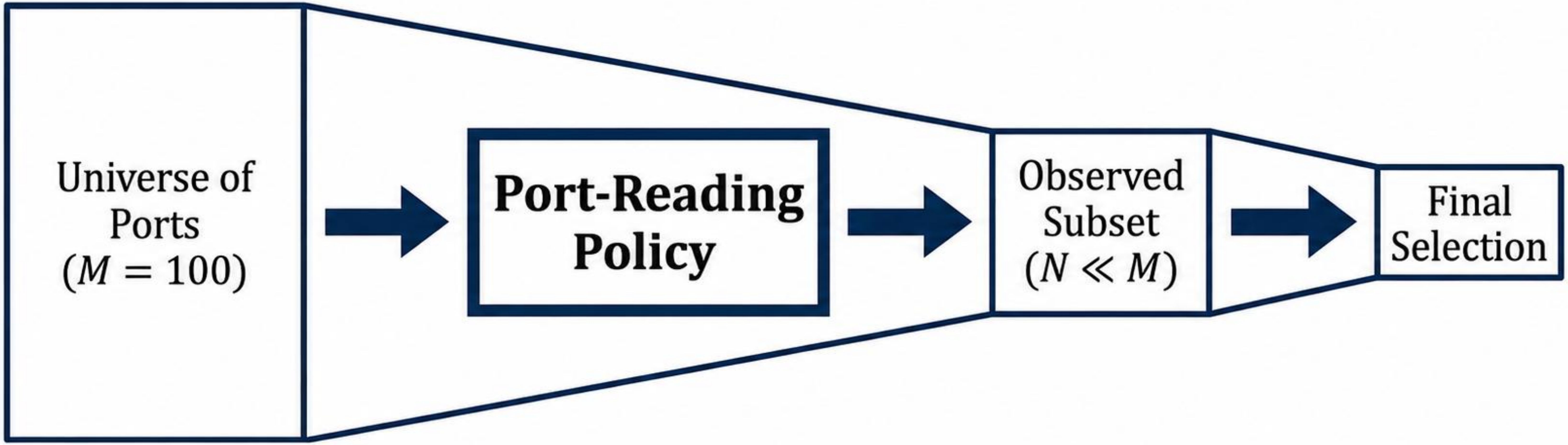
The Bottleneck



Reading all M ports induces severe measurement overhead, switching delay, and heavy receiver complexity.

THE RESEARCH GAP: Current machine learning approaches focus on selecting ports after all measurements are taken. But what if strict latency constraints mean we can only afford to measure a tiny fraction?

Isolating the Sensing Pattern from Downstream Selection



Core Objective

Treat port reading as a distinct policy-level design problem for measurement-limited FAS.

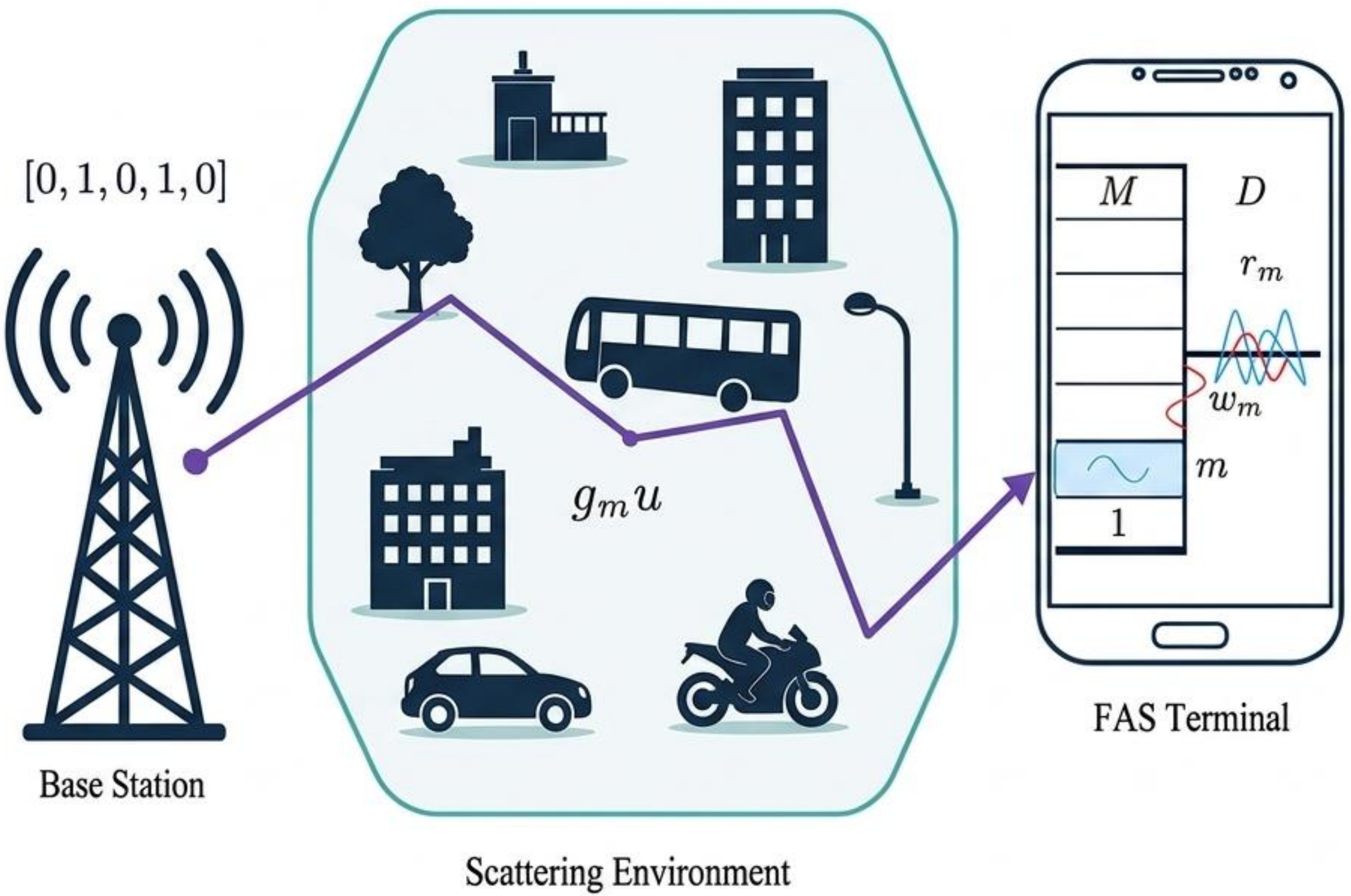
Key Question

Should the measured set prioritize geometric coverage, marginal port quality, or complementarity?

Evaluation Metric

Observed-only outage probability. The receiver selects strictly from the measured ports, isolating the efficacy of the reading policy from downstream reconstruction algorithms.

System Model: Single-Antenna BS to a Multi-Port FAS Terminal



Mathematical Constraints

M: Candidate ports over normalized aperture $D = L/\lambda$

$r_m = g_m u + w_m$: Baseband sample at port m

η: Full Signal-to-Noise Ratio (SNR) profile

$\mathcal{O}$: Index set of measured ports ($N \ll M$)

The receiver operates strictly from observation vector $\eta_{\mathcal{O}}$, never the full profile η.

Spatially Correlated κ–μ Shadowed Fading Environment

Correlation & Power Models

Sub-wavelength spacing forces high correlation, modeled via a Jakes-type matrix:

$$[R]_{m,n} = J_0(2\pi|m-n|/D(M-1)).$$

Power Model:

$$|g_m|^2 = \sum_{l=1}^{\mu} |q_{m,l} + p_m|^2.$$

p_m	Dominant Component	Perfectly correlated, shifts entire aperture coherently.
$q_{m,l}$	Scattered Components	The primary source of spatial diversity.

Dataset Regimes

Generated 14 diverse fading regimes manipulating dominant-to-scattered power (κ), clusters (μ), and shadowing severity (m).

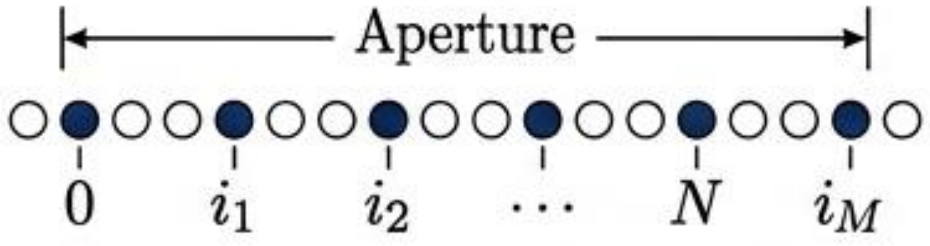
Severe Rayleigh Shadowing

Strong Rician Conditions

Ensures evaluation robustness across heterogeneous real-world conditions.

Three Approaches to the Port-Reading Problem

Policy A (Uniform Reading)



Method: Subsamples the aperture by evenly spacing N indices.

Focus: Geometric spatial coverage. Deterministic and requires no data-driven tuning.

Policy B (Marginal Ranking)

Method: Scores and ranks ports based on historical exceedance:

$$\hat{p}_m = \frac{1}{|\mathcal{D}_{des}|} \sum b_{i,m}$$

Focus: Threshold-aware per-port individual quality.

Policy C (Greedy Coverage)

Method: Iteratively adds ports that maximize the number of newly covered samples.

Focus: Complementarity and set-level diversity.

Rigorous Monte Carlo Evaluation Protocol

$S = 10^6$ samples/regime	$M = 100$ ports	Budgets $N \in \{3, 4, 5, 6, 7\}$	Target Threshold $\eta_{\text{th}} = 0.8$

Algorithm 1: Monte Carlo Evaluation

1. Split Data

Divide 1 million samples into Design ($\mathcal{D}_{des}$) and Evaluation ($\mathcal{D}_{eval}$) subsets.

2. Learn

Train Policies B and C on $\mathcal{D}_{des}$ using historical exceedance events.

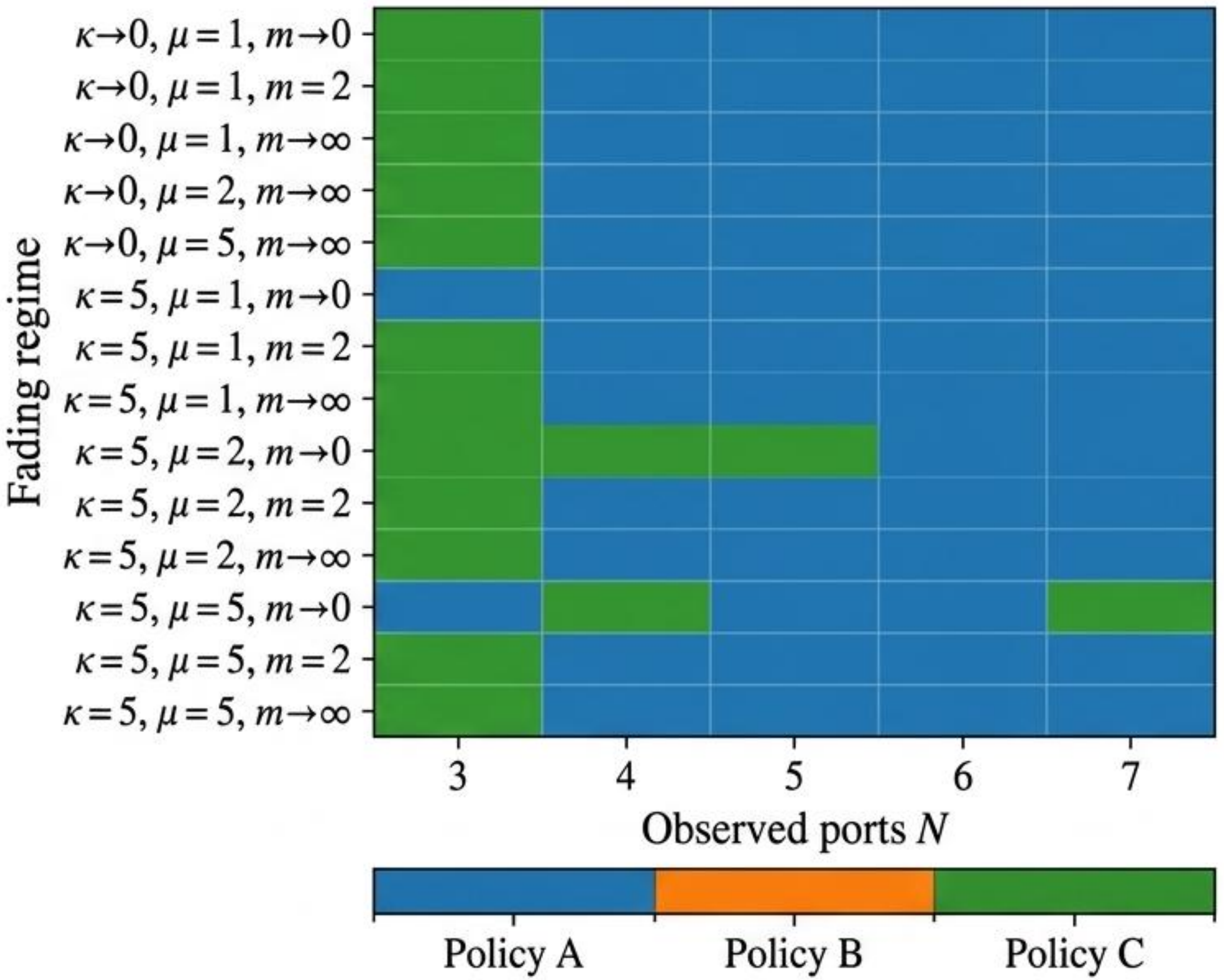
3. Evaluate

Test Policies A, B, and C for observed-only outage purely on unseen $\mathcal{D}_{eval}$.

4. Repeat

Execute 100 independent splits per regime to guarantee statistical stability.

The Budget-Dependent Crossover Effect



Tight Budget ($N = 3$)

Policy C (Greedy Coverage) wins in 12 of 14 regimes (85.7%). When measurement starved, explicitly searching for complementarity is optimal.

Relaxed Budget ($N \geq 4$)

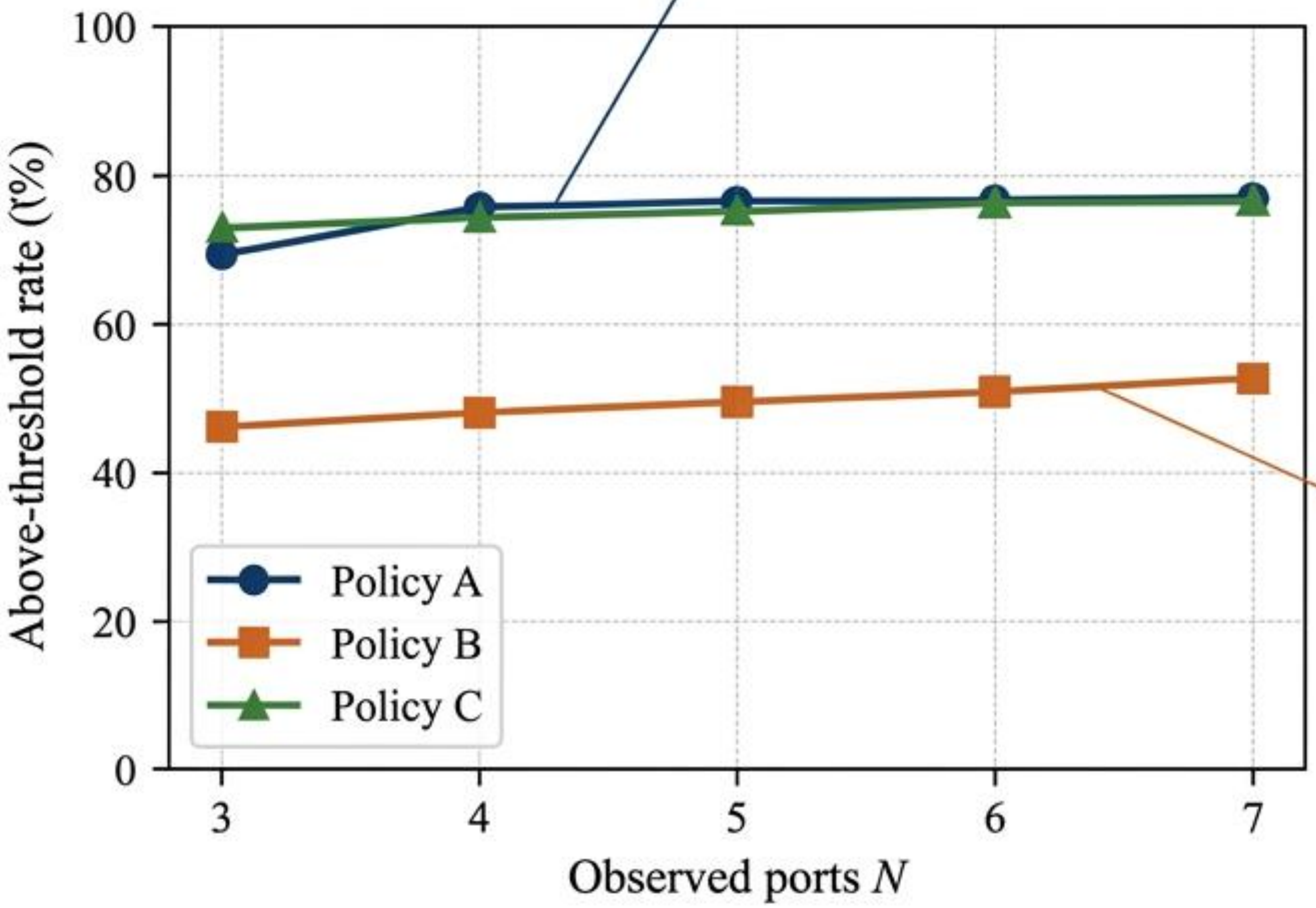
Policy A (Uniform Subsampling) completely dominates, hitting a 100% win rate at $N = 6$. A single additional observation fundamentally shifts the optimal strategy.

The Anomaly

Policy B (Marginal Ranking) achieves a 0% win rate across all regimes and budgets.

System Reliability vs. Observation Budget

1. Convergence: Policies A and C rapidly approach ideal theoretical performance by N=5.



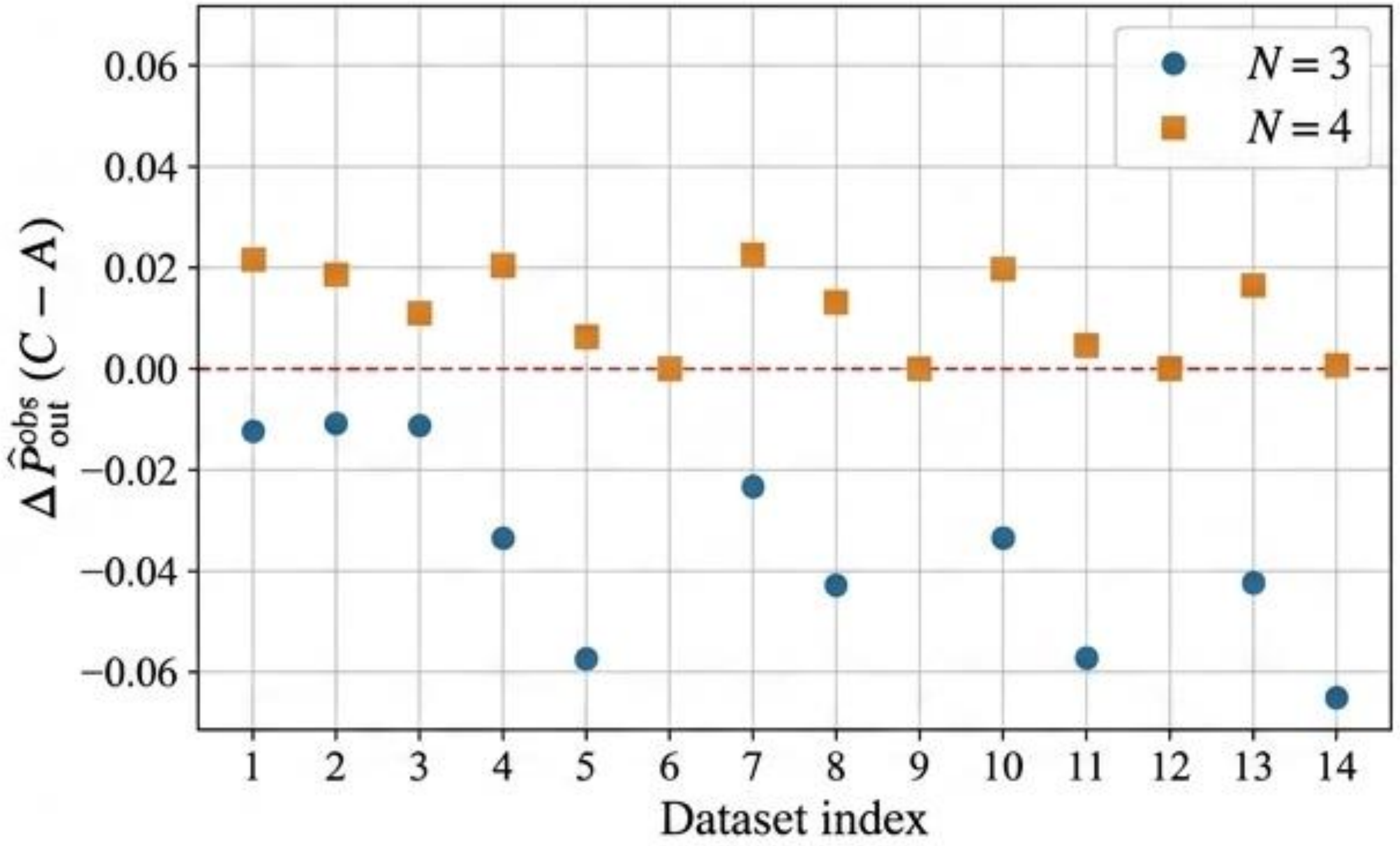
2. The Failure of Policy B: Even with a budget of 7 ports, selecting historically "best" individual ports severely lags behind basic uniform sampling. High individual exceedance frequency is a poor proxy for set-level diversity.

Statistical Certainty of the N=3 to N=4 Crossover

Blue Dots ($N = 3$):

Consistently negative across datasets.

Greedy coverage is statistically superior under extreme starvation.



Orange Squares ($N = 4$):

Consistently positive across datasets.

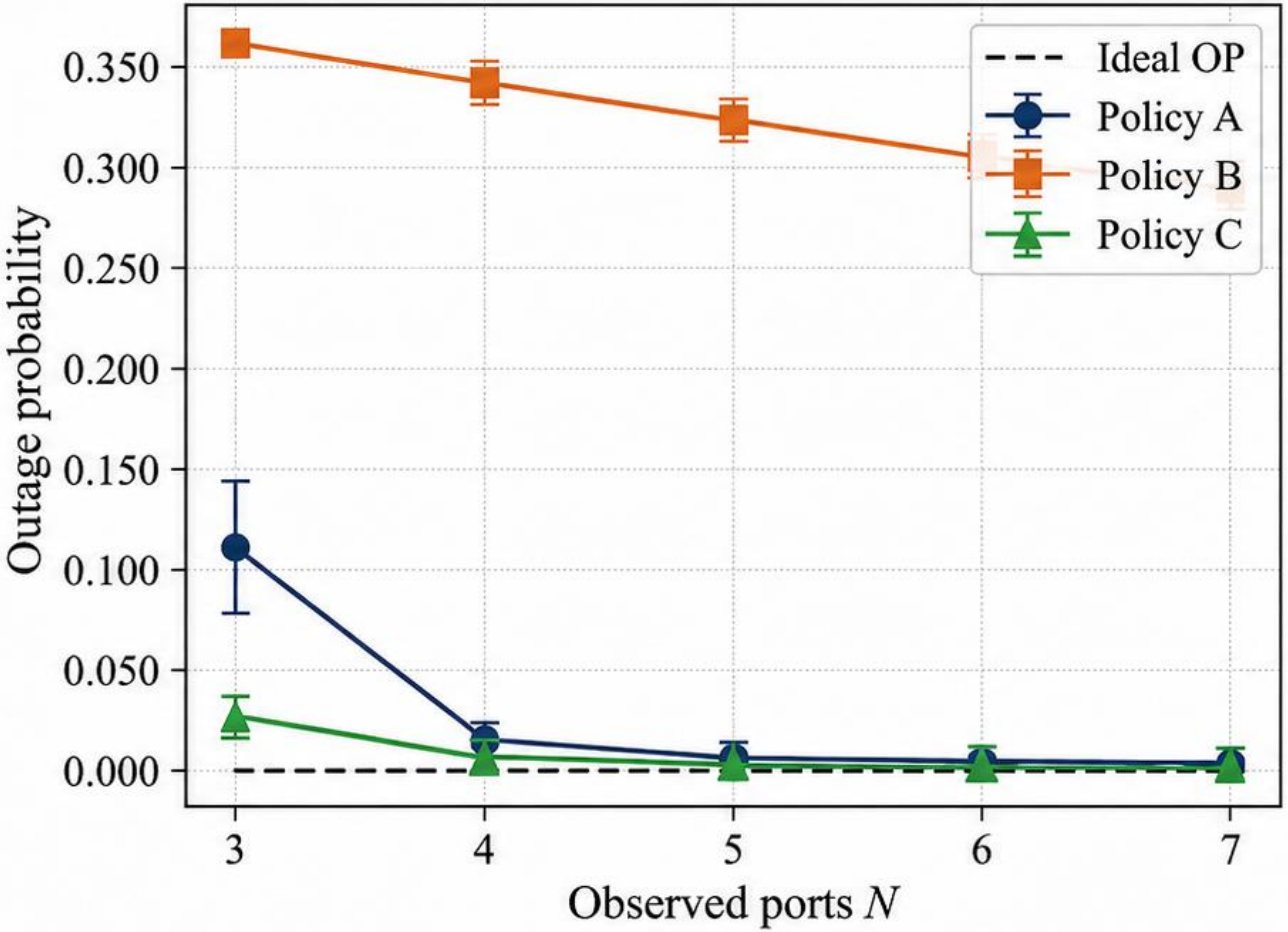
The addition of a single observation fundamentally shifts the optimal strategy to **Uniform** coverage.

Metric Definition: $\Delta \widehat{P}^{obs}_{out} = C - A$

Negative values favor **Policy C.**
Positive values favor **Policy A.**

Approaching Ideal Performance Limits

(Scattered-field dominated regime: $\kappa \to 0, \mu = 5, m \to \infty$)



Theoretical Floor

The black dashed line represents the Ideal Outage Probability, the absolute minimum possible outage if all 100 ports were measured.

High Efficiency

Policies A and C drive the outage probability down to near-ideal levels with just 4 to 5 measurements.

Absolute Stability

The 95% confidence intervals are smaller than the data markers, verifying the stability of the reading policies across 100 independent splits.

The Computational Reality: Selection-Time Benchmark

Policy A (Uniform)

0.0638 ms

Mean Selection Time. Requires zero data-driven tuning. Instantaneous operation.

Policy C (Greedy)

1,595.63 ms

Mean Selection Time. Requires iterating through a massive design dataset.

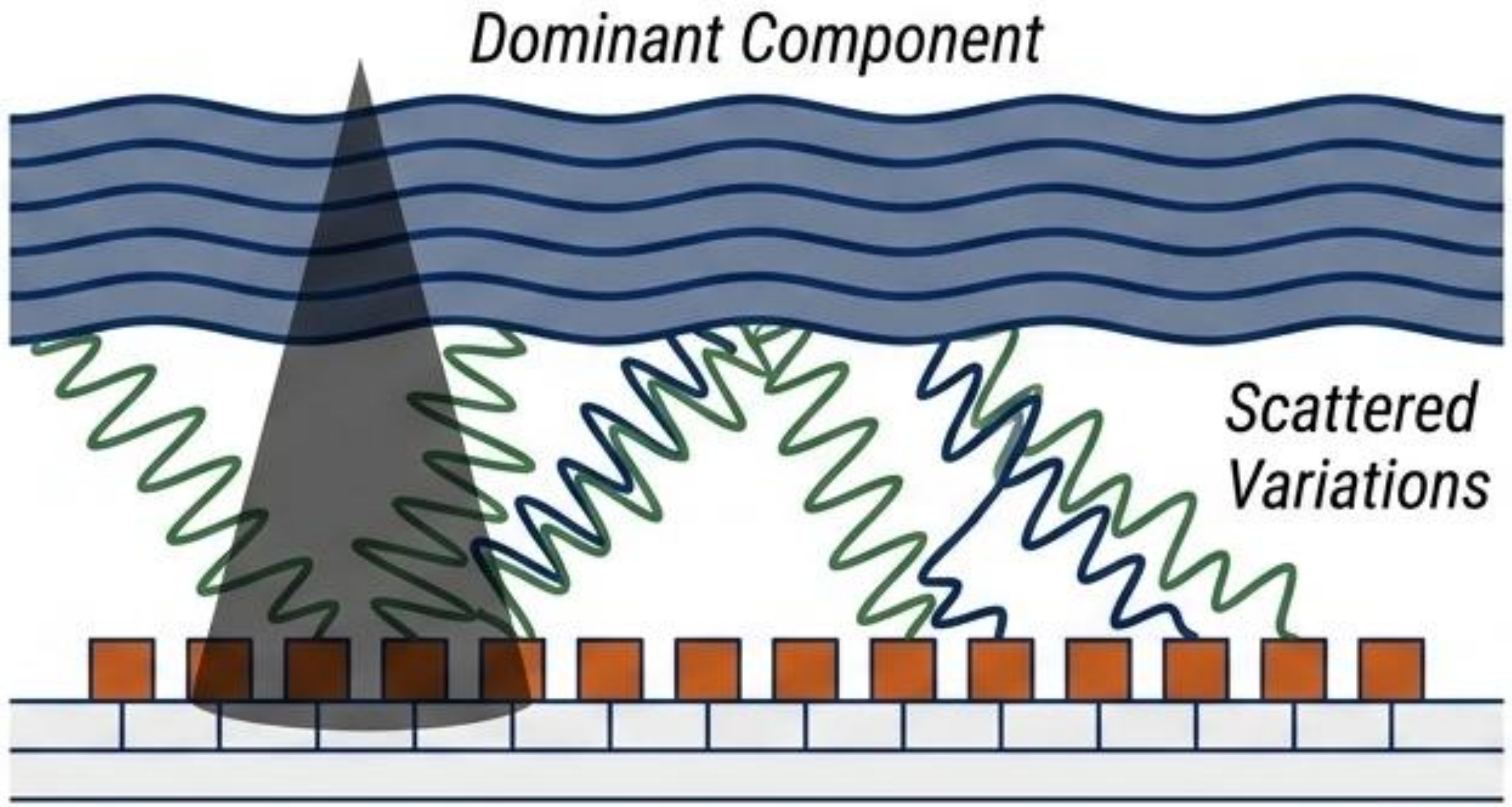
THE VERDICT: Policy C is approximately 25,000x slower. For any budget $N \geq 4$, Policy A delivers optimal outage performance with practically instantaneous selection.

The Physics of the Aperture: Why Marginal Ranking Fails

The Mechanism

The dominant signal component shifts the entire aperture coherently. It does not create spatial diversity.

Spatial diversity is generated entirely by sub-wavelength scattered field variations.



The Synthesis

Because all ports share similar marginal statistics, ranking them in isolation (Policy B) simply selects a clustered group of physically proximate ports. If a shadow drops on that cluster, all selected ports fail together.

A system must either blindly sample across the entire spatial variation (Policy A) or explicitly compute complementary variations (Policy C).

Establishing Port Reading as a FAS Design Layer

Recommendation 1: Extreme Constraints ($N \leq 3$)

Use Greedy Coverage. When strictly limited by hardware or latency, paying the computational cost explicitly secures complementary ports and minimizes observed outage.

Recommendation 2: Operational Default ($N \geq 4$)

Default to Uniform Reading. It is highly robust, converges rapidly to ideal theoretical performance limits, and avoids massive computational training overhead.

Recommendation 3: Avoid Marginal Selection

Do not rank ports by historical individual quality. Per-port individual success frequency does not equate to set-level spatial diversity due to severe sub-wavelength correlation.

Future Directions & Acknowledgments

Future Research Vectors

Varying Aperture Lengths ($D = 0.5$, $D = 2$) to test the boundary limits of spatial correlation.

Correlation-aware subset selection (e.g., maximizing covariance determinant) to achieve Policy C performance without training samples.

Joint sensing-policy optimization under strict hardware latency and switching constraints.

Acknowledgments

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CNPq xGMobile RNP FAPEMIG FINEP

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